

6.1 The Poisson distribution

Probability and Statistics 2 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Poisson models counts of events in a fixed interval under independent, constant-rate assumptions. $P(X=r)=e^{-\lambda} \lambda^r/r!$, with mean and variance λ . For twice the exposure, double λ ; a binomial approximation may apply for large n and small p with $\lambda=np$.

Worked example: follow the reasoning

If mean calls per hour is 2, probability none in one hour is $e^{-2} \approx 0.1353$.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Calls arrive at mean rate 3 per hour. Find $P(\text{no calls in 20 minutes})$.

[3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

Mean for $1/3$ h is $\lambda=1$ [1]; $P(X=0)=e^{-1} 1^0/0!$ [1] ≈ 0.3679 [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.